Saad Abdur Razzaq

Machine Learning Researcher | *Optimization & Training Dynamics* | *Computer Vision*
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LinkedIn | GitHub | YouTube | The Applied Engineer SubStack

Personal Summary

Machine learning researcher interested in optimization, training dynamics, and the mechanisms underlying learning and generalization in deep neural networks. My work involves implementing learning algorithms and neural architectures from scratch, systematically comparing model architectures, and analyzing computational and predictive efficiency, with a particular focus on computer vision and visual representation learning. I design controlled experiments to investigate model behavior, test hypotheses, quantify performance differences, and communicate findings through reproducible analysis and visualization.

Research Interests

Core: Deep Learning Optimization, Training Dynamics, Architecture Analysis, Model Efficiency, Computer Vision
Methods: From-Scratch Implementation, Systematic Comparison, Quantified Evaluation, Visualization

Education

National University of Computer and Emerging Sciences (FAST–NUCES) *Sep 2020 – Jun 2024*
B.Sc. Computer Science Chiniot–Faisalabad, Pakistan

- Relevant coursework: Artificial Intelligence, Fundamentals of Computer Vision, Applied Machine Learning, Numerical Computing, Probability and Statistics

Research Experience

Adaptive RMSprop+: Enhanced Optimization Framework for Deep Learning GitHub

- Designed and implemented a novel extension of the RMSprop optimizer incorporating dynamic epsilon adjustment, cyclical learning rates with warmup, adaptive gradient noise, and layer-wise momentum adaptation.
- Achieved 28% faster convergence, 60% improvement in training stability, and 1.5% absolute accuracy gain over standard RMSprop across image classification benchmarks.
- Implemented the optimizer from scratch in PyTorch, including mathematical formulation of gradient-norm-aware epsilon scaling and cosine-annealed cyclical learning rate schedules.
- Conducted systematic comparison against Adam and standard RMSprop, demonstrating superior convergence across CIFAR-10 and language modeling tasks.

From-Scratch Recurrent Neural Network for Sequential Arithmetic Computation GitHub

- Implemented a vanilla RNN entirely from first principles using NumPy, including manual forward propagation, backpropagation through time, gradient clipping, and numerical gradient checking.
- Investigated the ability of recurrent architectures to learn symbolic arithmetic operations (cumulative products) through temporal dependencies, demonstrating successful convergence on sequential computation tasks.
- Validated implementation correctness through numerical gradient checking, ensuring analytical gradients matched numerical approximations to high precision.
- Analyzed training dynamics, gradient propagation, and weight initialization to understand how RNNs represent and learn sequential computation.

Memory-Augmented Reinforcement Learning: A Systematic Comparison Framework GitHub

- Built a modular experimental framework for systematically comparing RL agents with and without explicit memory mechanisms across multiple decision-making tasks.
- Implemented memory-augmented Q-learning and DQN agents from scratch to isolate the impact of memory on learning dynamics.
- Demonstrated 63% performance improvement with memory augmentation in T-Maze task and 56% in spatial alternation, with evidence of cross-task transfer benefits.
- Developed visualization and analytics tools for learning curves, memory utilization patterns, and transfer learning analysis.

Quantization and Efficiency Analysis of Transformer-Based Sequence Models GitHub

- Built an experimental framework for systematically evaluating the efficiency-quality trade-off in transformer-based neural machine translation under INT8 quantization.
- Quantized Helsinki-NLP OPUS-MT model using ONNX Runtime, achieving 4× model size reduction, 2.7× inference latency improvement, and 4× memory reduction.
- Maintained translation quality within 1.2% BLEU score of the original FP32 model, analyzing layer sensitivity & calibration strategies for preserving quality under aggressive compression.

Robustness Evaluation of FaceNet-Based Recognition Systems GitHub

- Developed a modular facial recognition pipeline using PyTorch, MTCNN detection, and FaceNet/InceptionResNetV1 embeddings.
- Evaluated model robustness under systematic distribution shifts including brightness variation, expression changes, image quality degradation, and age variance.
- Achieved 95.2% accuracy on standard conditions and systematically analyzed failure modes through confusion matrices, ROC curves, and confidence calibration.

Selected Projects

Synthetic Sequence Generation with LSTM Networks GitHub

- Developed an LSTM-based sequence-to-sequence framework for generating realistic synthetic sequences from structured sequential data.
- Learned concept embeddings using skip-gram architecture with negative sampling to capture semantic relationships between features.

Deep Q-Network for CartPole GitHub

- Implemented DQN from scratch in PyTorch with experience replay and epsilon-greedy exploration to solve CartPole-v1.
- Achieved convergence within 200–500 episodes, demonstrated stable learning via replay buffer & target network updates.

Final Year Project

GynaeAutoCare – Digital Healthcare Platform for Women's Health *FAST–NUCES*

- Collected and prepared real-world clinical data from local hospitals and developed a machine learning pipeline using XGBoost for data-driven clinical prediction.
- Applied feature engineering and model evaluation to transform real-world patient data into predictive insights for clinical decision support.

Industry Experience

Software Developer (Full Stack) *July 2024 – Present*
ProActive Enterprises (Private) Limited Remote, Pakistan

- Developed and maintained full-stack applications across frontend, backend, and database layers, implementing authentication, API integrations, and database-driven workflows for production systems.

Teaching Assistant – Data Science & Business Analytics *Aug 2023 – Jun 2024*
FAST–NUCES, Chiniot–Faisalabad Campus Pakistan

- Supported student learning in Python-based data analysis, machine learning, & predictive modeling through tutorials, technical guidance, & code reviews; helped students debug implementations & develop high-quality solutions.

Technical Skills

ML Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, LightGBM, ONNX Runtime, FastAI
Architectures: CNNs, RNNs, LSTMs, DQN, VGG, FaceNet, Vision Transformers (ViT), Seq2Seq, RestNET
Techniques: Optimization Algorithms, Model Quantization, Transfer Learning, Knowledge Distillation, Regularization
Languages: Python, C, C++, R, SQL
Tools: NumPy, Pandas, OpenCV, Git, Docker, PostgreSQL, FastAPI

Selected Certifications

- IBM – AI Developer Professional Certificate, 2025 Verify
- IBM – Generative AI Engineering Professional Certificate, 2025 Verify
- Google – Advanced Data Analytics Professional Certificate, 2025 Verify

Knowledge Sharing

Founder & Author – The Applied Engineer Substack

- Authoring technical articles that explain the mathematical foundations of machine learning, including loss functions, gradients, backpropagation, and training dynamics, for readers without prior ML background.
- Developing intuitive, example-driven explanations that connect mathematical concepts to practical machine learning workflows and model behavior.
- Translating concepts from optimization, neural network training, and model internals into accessible technical content while maintaining mathematical rigor.

Languages

English: IELTS Academic 7.0 overall (no band below 6.5) | **Urdu:** Native